

# AAMOD VARMA

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## EDUCATION

### Georgia Institute of Technology

Atlanta, GA

*BS in Computer Science and BS in Mathematics, GPA: 4.0*

*May 2027*

**Relevant Coursework:** Machine Learning, Deep Learning, DSA, Computer Hardware, OS, Compilers, Analysis

**Awards:** 1st place at Hacklytics, Georgia Tech - Traversaal AI Challenge (2024); UAE National Champions, F1 in Schools (2021-22); 1st place at Abu Dhabi University Programming Competition, UAE (2021)

## PUBLICATIONS

- Candocia, A., Inan, O. M., Agarwal, R., **Varma, A.**, Davenport, M. A. (2025). Scalable continual learning: Adaptive MoEs for expanding task sets. In *Proceedings of the ICLR 2025 Workshop on Sparsity in LLMs (SLLM)*. Short paper.
- Jia, S., **Varma, A.**, Manivannan, P., Chayapathy, D., Fung, V. (2025). Benchmarking text representations for crystal structure generation with large language models. In *Proceedings of ICLR 2025* (Conference paper).

## WORK EXPERIENCE

### PRGX, Atlanta

Jan 2025 – Jul 2025

*AI Intern*

*Atlanta, GA*

- Optimized table extraction and bounding box detection via fine-tuned Qwen VLM + PaddleOCR pipeline, reducing document processing time by 90% and TEDS from 0.5 to 0.8.
- Built evaluation framework using TEDS, Recall, and BLEU to assess model performance.
- Implemented NER for key information extraction using a fine-tuned BERT and Mistral model, boosting document parsing accuracy.

## RESEARCH AND PROJECTS

### Mark Davenport Research Group, Georgia Tech

July 2024 – Present

*Undergraduate Research Assistant*

*Atlanta, GA*

- Investigated continual learning approaches using Mixture of Expert models to handle evolving task distributions.
- Developed an evaluation framework for custom subspace-based dataset, MNIST and CIFAR-10/100, within a continual learning task stream.
- Developed covariance-based method to classify Gaussian tasks in Mixture of Experts (MOE) using Mahalanobis distance and log-determinant scoring.

### Fung Group, Georgia Tech

Oct 2023 – Present

*Undergraduate Research Assistant*

*Atlanta, GA*

- Implemented diffusion-based pretraining for an equivariant GNN (Orb-v2) on 60K Materials Project structures within the MatterTune framework, improving downstream property-prediction MAE by 30% over random initialization across three Matbench tasks.
- Built and compared discrete variance-exploding and continuous VP/sub-VP score-based formulations with timestep-conditioned denoising heads, isolating the noise schedule's effect on representation quality.
- Developed three synthetic equilibrium-structure generation pipelines, MACE-relaxed perturbations, MatterGen sampling, and protostructure enumeration via Latin hypercube sampling, expanding the pretraining set 2-3x to study data-scaling effects.

### Structured-Mask Diffusion for Image Restoration

May 2025 – Jul 2025

- Built mask-conditioned and mask-weighted DDPM variants (U-Net backbone, 1000-step diffusion) on FFHQ faces, reducing FID from 48.0 to 20.3 by injecting edge and semantic masks as input conditioning channels.
- Implemented and compared three spatial-prior strategies: mask conditioning, mask-weighted loss, and adaptive noise injection—isolating conditioning as the only consistent gain and showing adaptive noise degrades performance by breaking the Gaussian forward-process assumption.

## SKILLS

**Languages:** Python, Java, C/C++, Bash, SQL, HTML/CSS, R

**Tools:** PyTorch, Django, Pandas, JUnit, Git, NumPy, Linux, L<sup>A</sup>T<sub>E</sub>X

**Soft Skills:** Communication, Teamwork, Problem-Solving, Adaptability, Leadership